

Courant Algebroid Relations and T-duality

Tom De Fraja

Heriot-Watt University
Supervisor: Richard Szabo

Outline

- 1 Sigma-models and Courant algebroids
- 2 T-duality and CA Relations
- 3 Applications
- 4 Outlook

Courant algebroids

The **generalised tangent bundle**, $TM \oplus T^*M$, can be equipped with a pairing and a bracket, $H \in \Omega_{cl}^3(M)$:

$$\begin{aligned}\langle X + \xi, Y + \eta \rangle &= \iota_X \eta + \iota_Y \xi, \\ [X + \xi, Y + \eta]_H &= [X, Y] + \mathcal{L}_X \eta - \iota_Y d\xi + \iota_X \eta.\end{aligned}$$

In general, an **exact Courant algebroid** (CA) is a vector bundle E over M with a pairing and a bracket of sections fitting the sequence

$$0 \rightarrow T^*M \rightarrow E \rightarrow TM \rightarrow 0. \quad (E \cong TM \oplus T^*M)$$

CAs are classified by their *Ševera class* $[H] \in H_{cl}^3(M)$, with H given by a splitting $s: TM \rightarrow E$

$$H(X, Y, Z) = \langle [s(X), s(Y)]_E, s(Z) \rangle_E.$$

Courant algebroids

The **generalised tangent bundle**, $TM \oplus T^*M$, can be equipped with a pairing and a bracket, $H \in \Omega_{cl}^3(M)$:

$$\begin{aligned}\langle X + \xi, Y + \eta \rangle &= \iota_X \eta + \iota_Y \xi, \\ [X + \xi, Y + \eta]_H &= [X, Y] + \mathcal{L}_X \eta - \iota_Y d\xi + \iota_X d\eta.\end{aligned}$$

In general, an **exact Courant algebroid** (CA) is a vector bundle E over M with a pairing and a bracket of sections fitting the sequence

$$0 \rightarrow T^*M \rightarrow E \rightarrow TM \rightarrow 0. \quad (E \cong TM \oplus T^*M)$$

CAs are classified by their *Ševera class* $[H] \in H_{cl}^3(M)$, with H given by a splitting $s: TM \rightarrow E$

$$H(X, Y, Z) = \langle [s(X), s(Y)]_E, s(Z) \rangle_E.$$

Sigma-models in generalised geometry

Consider maps f from a worldsheet surface Σ to a target space M .

Sigma-Model	Geometry
Wess-Zumino term $S_H[f] = \int_V f^* H$	Courant algebroid E_H over M
$\delta S_H = 0$	$X + \xi$ such that $\iota_X H = d\xi$
Polyakov term $S_g[f] = \int_\Sigma \ df\ _g^2 d\mu_g$	Generalised metric $V_+ \subset E_H$ equiv. $\tau \in \text{End}(E_H)$, $\tau^2 = 1$ equiv. $g \in \text{Riem}(M)$
$\delta S_g = 0$	$\mathcal{L}_X g = 0$

The data of the full action $S = S_H + S_g$ is encoded in a CA equipped with a positive definite subbundle, (E, V_+) .

Sigma-models in generalised geometry

Consider maps f from a worldsheet surface Σ to a target space M .

Sigma-Model	Geometry
Wess-Zumino term $S_H[f] = \int_V f^* H$	Courant algebroid E_H over M
$\delta S_H = 0$	$X + \xi$ such that $\iota_X H = d\xi$
Polyakov term $S_g[f] = \int_\Sigma \ df\ _g^2 d\mu_g$	Generalised metric $V_+ \subset E_H$ equiv. $\tau \in \text{End}(E_H)$, $\tau^2 = 1$ equiv. $g \in \text{Riem}(M)$
$\delta S_g = 0$	$\mathcal{L}_X g = 0$

The data of the full action $S = S_H + S_g$ is encoded in a CA equipped with a positive definite subbundle, (E, V_+) .

Sigma-models in generalised geometry

Consider maps f from a worldsheet surface Σ to a target space M .

Sigma-Model	Geometry
Wess-Zumino term $S_H[f] = \int_V f^* H$	Courant algebroid E_H over M
$\delta S_H = 0$	$X + \xi$ such that $\iota_X H = d\xi$
Polyakov term $S_g[f] = \int_\Sigma \ df\ _g^2 d\mu_g$	Generalised metric $V_+ \subset E_H$ equiv. $\tau \in \text{End}(E_H)$, $\tau^2 = 1$ equiv. $g \in \text{Riem}(M)$
$\delta S_g = 0$	$\mathcal{L}_X g = 0$

The data of the full action $S = S_H + S_g$ is encoded in a CA equipped with a positive definite subbundle, (E, V_+) .

Sigma-models in generalised geometry

Consider maps f from a worldsheet surface Σ to a target space M .

Sigma-Model	Geometry
Wess-Zumino term $S_H[f] = \int_V f^* H$	Courant algebroid E_H over M
$\delta S_H = 0$	$X + \xi$ such that $\iota_X H = d\xi$
Polyakov term $S_g[f] = \int_\Sigma \ df\ _g^2 d\mu_g$	Generalised metric $V_+ \subset E_H$ equiv. $\tau \in \text{End}(E_H)$, $\tau^2 = 1$ equiv. $g \in \text{Riem}(M)$
$\delta S_g = 0$	$\mathcal{L}_X g = 0$

The data of the full action $S = S_H + S_g$ is encoded in a CA equipped with a positive definite subbundle, (E, V_+) .

Courant algebroid relations

Isomorphism $\Phi: E_1 \rightarrow E_2$ of CAs covering a diffeomorphism $\varphi: M_1 \rightarrow M_2$ preserve all CA structure, hence the Sigma-models over M_1 and M_2 have the same equations of motion.

T-duality is an equivalence of Sigma-models on different target manifolds M_1 and M_2 , though these are not necessarily diffeomorphic. A CA isomorphism $\Phi: E_1 \rightarrow E_2$ must cover a diffeomorphism, so we seek to generalise the notion of isomorphism.

Consider the graph $\text{gr}(\Phi) \subset E_1 \times E_2$. One sees that

$$[\Phi \cdot, \Phi \cdot]_2 = \Phi([\cdot, \cdot]_1) \iff \text{gr}(\Phi) \text{ is involutive in } E_1 \times E_2$$

$$\langle \Phi \cdot, \Phi \cdot \rangle_2 = \langle \cdot, \cdot \rangle_1 \iff \text{gr}(\Phi) \text{ is isotropic in } E_1 \times E_2$$

Courant algebroid relations

Isomorphism $\Phi: E_1 \rightarrow E_2$ of CAs covering a diffeomorphism $\varphi: M_1 \rightarrow M_2$ preserve all CA structure, hence the Sigma-models over M_1 and M_2 have the same equations of motion.

T-duality is an equivalence of Sigma-models on different target manifolds M_1 and M_2 , though these are not necessarily diffeomorphic. A CA isomorphism $\Phi: E_1 \rightarrow E_2$ must cover a diffeomorphism, so we seek to generalise the notion of isomorphism.

Consider the graph $\text{gr}(\Phi) \subset E_1 \times E_2$. One sees that

$$[\Phi \cdot, \Phi \cdot]_2 = \Phi([\cdot, \cdot]_1) \iff \text{gr}(\Phi) \text{ is involutive in } E_1 \times E_2$$

$$\langle \Phi \cdot, \Phi \cdot \rangle_2 = \langle \cdot, \cdot \rangle_1 \iff \text{gr}(\Phi) \text{ is isotropic in } E_1 \times E_2$$

Courant algebroid relations

Isomorphism $\Phi: E_1 \rightarrow E_2$ of CAs covering a diffeomorphism $\varphi: M_1 \rightarrow M_2$ preserve all CA structure, hence the Sigma-models over M_1 and M_2 have the same equations of motion.

T-duality is an equivalence of Sigma-models on different target manifolds M_1 and M_2 , though these are not necessarily diffeomorphic. A CA isomorphism $\Phi: E_1 \rightarrow E_2$ must cover a diffeomorphism, so we seek to generalise the notion of isomorphism.

Consider the graph $\text{gr}(\Phi) \subset E_1 \times E_2$. One sees that

$$[\Phi \cdot, \Phi \cdot]_2 = \Phi([\cdot, \cdot]_1) \iff \text{gr}(\Phi) \text{ is involutive in } E_1 \times E_2$$

$$\langle \Phi \cdot, \Phi \cdot \rangle_2 = \langle \cdot, \cdot \rangle_1 \iff \text{gr}(\Phi) \text{ is isotropic in } E_1 \times E_2$$

Isometry and Composition

A **CA relation** $R: E_1 \dashrightarrow E_2$ is an isotropic, involutive subbundle of $E_1 \times E_2$, supported on a submanifold $C \subset M_1 \times M_2$.

If $R: (E_1, \tau_1) \dashrightarrow (E_2, \tau_2)$, then R is a **generalised isometry** if

$$(\tau_1 \times \tau_2)(R) = R.$$

Contingent on some smoothness conditions [Vysoký, 1], one can compose two relations $R: E_1 \dashrightarrow E_2$, $\tilde{R}: E_2 \rightarrow E_3$

$$\tilde{R} \circ R = \{(e_1, e_3) : (e_1, e_2) \in R \text{ and } (e_2, e_3) \in \tilde{R}\} \subset E_1 \times \overline{E}_3,$$

Isometry and Composition

A **CA relation** $R: E_1 \dashrightarrow E_2$ is an isotropic, involutive subbundle of $E_1 \times E_2$, supported on a submanifold $C \subset M_1 \times M_2$.

If $R: (E_1, \tau_1) \dashrightarrow (E_2, \tau_2)$, then R is a **generalised isometry** if

$$(\tau_1 \times \tau_2)(R) = R.$$

Contingent on some smoothness conditions [Vysoký, 1], one can compose two relations $R: E_1 \dashrightarrow E_2$, $\tilde{R}: E_2 \rightarrow E_3$

$$\tilde{R} \circ R = \{(e_1, e_3) : (e_1, e_2) \in R \text{ and } (e_2, e_3) \in \tilde{R}\} \subset E_1 \times \bar{E}_3,$$

Isometry and Composition

A **CA relation** $R: E_1 \dashrightarrow E_2$ is an isotropic, involutive subbundle of $E_1 \times E_2$, supported on a submanifold $C \subset M_1 \times M_2$.

If $R: (E_1, \tau_1) \dashrightarrow (E_2, \tau_2)$, then R is a **generalised isometry** if

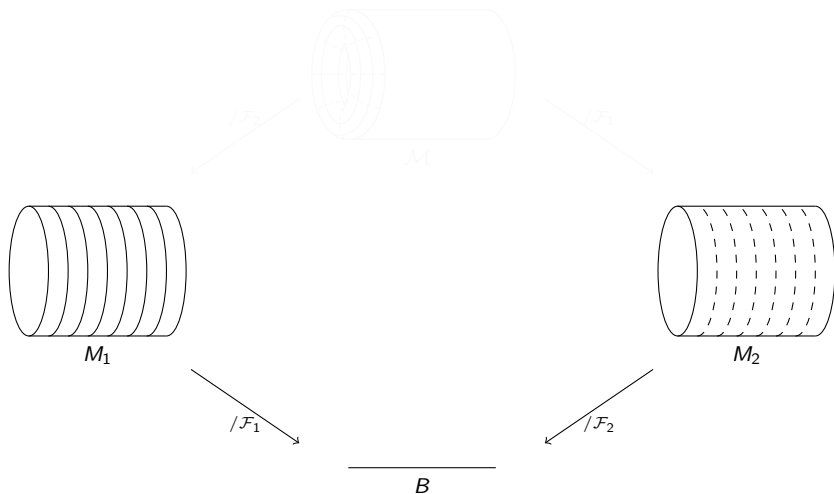
$$(\tau_1 \times \tau_2)(R) = R.$$

Contingent on some smoothness conditions [Vysoký, 1], one can compose two relations $R: E_1 \dashrightarrow E_2$, $\tilde{R}: E_2 \rightarrow E_3$

$$\tilde{R} \circ R = \{(e_1, e_3) : (e_1, e_2) \in R \text{ and } (e_2, e_3) \in \tilde{R}\} \subset E_1 \times \overline{E}_3,$$

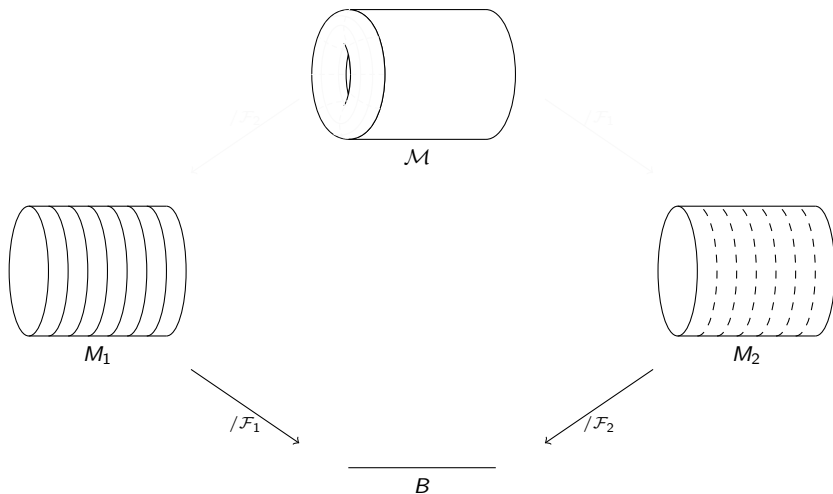
T-duality

T-dual spaces M_1, M_2 may be fibre bundles over a common base B



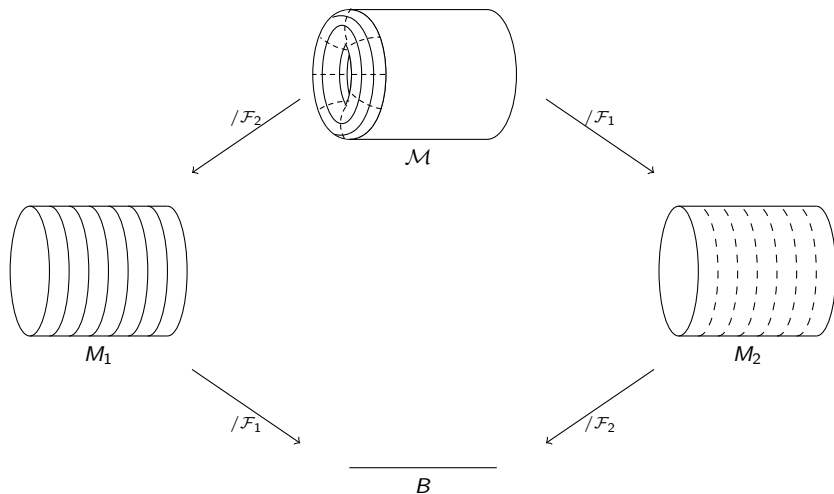
T-duality

T-dual spaces M_1, M_2 may be fibre bundles over a common base B



T-duality

T-dual spaces M_1, M_2 may be fibre bundles over a common base B



The T-duality relation I

Want to form a CA relation $R: E_1 \dashrightarrow E_2$ between CAs E_i over M_i . Need to know how to form CAs on a quotient manifold:

$\mathcal{M}$ is foliated by $\mathcal{F}_2$. If $\mathcal{E}$ is an exact CA over $\mathcal{M}$, then $\mathcal{E}/\mathcal{F}_2$ will not be an exact CA over $M_1 = \mathcal{M}/\mathcal{F}_2$.

Take $K_2 = T\mathcal{F}_2 \subset \mathcal{E}$, then

$$E = \frac{K_2^\perp}{K_2} / \mathcal{F}_2$$

is an exact CA over M_1 [2,3].

Can form the CA relation $Q(K_2)$, supported on $\text{gr}(q_2) \subset \mathcal{M} \times M_1$,

$$Q(K_2) = \{(e, \natural_2(e)) : e \in K_2^\perp\} \subset \mathcal{E} \times E_1$$

where $\natural_2: K_2^\perp \rightarrow E_1$ is the quotient map [1].

The T-duality relation I

Want to form a CA relation $R: E_1 \dashrightarrow E_2$ between CAs E_i over M_i . Need to know how to form CAs on a quotient manifold: $\mathcal{M}$ is foliated by $\mathcal{F}_2$. If $\mathcal{E}$ is an exact CA over $\mathcal{M}$, then $\mathcal{E}/\mathcal{F}_2$ will not be an exact CA over $M_1 = \mathcal{M}/\mathcal{F}_2$.

Take $K_2 = T\mathcal{F}_2 \subset \mathcal{E}$, then

$$E = \frac{K_2^\perp}{K_2} / \mathcal{F}_2$$

is an exact CA over M_1 [2,3].

Can form the CA relation $Q(K_2)$, supported on $\text{gr}(q_2) \subset \mathcal{M} \times M_1$,

$$Q(K_2) = \{(e, \natural_2(e)) : e \in K_2^\perp\} \subset \mathcal{E} \times E_1$$

where $\natural_2: K_2^\perp \rightarrow E_1$ is the quotient map [1].

The T-duality relation I

Want to form a CA relation $R: E_1 \dashrightarrow E_2$ between CAs E_i over M_i . Need to know how to form CAs on a quotient manifold: $\mathcal{M}$ is foliated by $\mathcal{F}_2$. If $\mathcal{E}$ is an exact CA over $\mathcal{M}$, then $\mathcal{E}/\mathcal{F}_2$ will not be an exact CA over $M_1 = \mathcal{M}/\mathcal{F}_2$. Take $K_2 = T\mathcal{F}_2 \subset \mathcal{E}$, then

$$E = \frac{K_2^\perp}{K_2} / \mathcal{F}_2$$

is an exact CA over M_1 [2,3].

Can form the CA relation $Q(K_2)$, supported on $\text{gr}(q_2) \subset \mathcal{M} \times M_1$,

$$Q(K_2) = \{(e, \natural_2(e)) : e \in K_2^\perp\} \subset \mathcal{E} \times E_1$$

where $\natural_2: K_2^\perp \rightarrow E_1$ is the quotient map [1].

The T-duality relation I

Want to form a CA relation $R: E_1 \dashrightarrow E_2$ between CAs E_i over M_i . Need to know how to form CAs on a quotient manifold: $\mathcal{M}$ is foliated by $\mathcal{F}_2$. If $\mathcal{E}$ is an exact CA over $\mathcal{M}$, then $\mathcal{E}/\mathcal{F}_2$ will not be an exact CA over $M_1 = \mathcal{M}/\mathcal{F}_2$. Take $K_2 = T\mathcal{F}_2 \subset \mathcal{E}$, then

$$E = \frac{K_2^\perp}{K_2} / \mathcal{F}_2$$

is an exact CA over M_1 [2,3].

Can form the CA relation $Q(K_2)$, supported on $\text{gr}(q_2) \subset \mathcal{M} \times M_1$,

$$Q(K_2) = \{(e, \natural_2(e)) : e \in K_2^\perp\} \subset \mathcal{E} \times E_1$$

where $\natural_2: K_2^\perp \rightarrow E_1$ is the quotient map [1].

The T-duality relation II

The T-duality relation $R: E \rightarrow E'$ is then the composition of CA relations:

$$\begin{array}{ccc}
 \mathcal{E} & \xrightarrow{\Phi = \varphi_* \circ e^F} & \mathcal{E} \\
 Q(K_2)^T \uparrow \text{---} & & \text{---} \downarrow Q(K_1) \\
 E_1 & \xrightarrow{\quad R \quad} & E_2
 \end{array}$$

Theorem (DF, Marotta, Szabo [4])

Let g_1 be a Riemannian metric on M_1 . If $\mathcal{L}_X g = \mathcal{L}_X F = 0$, $\forall X \in \mathcal{F}_2$, then TFAE

- ① $K_1^\perp \cap \Phi(K_2) \subseteq K_1$.
- ② There exists a unique Riemannian metric g_2 on M_2 such that R is a generalised isometry.

The T-duality relation II

The T-duality relation $R: E \rightarrow E'$ is then the composition of CA relations:

$$\begin{array}{ccc}
 \mathcal{E} & \xrightarrow{\Phi = \varphi_* \circ e^F} & \mathcal{E} \\
 Q(K_2)^T \uparrow \text{---} & & \text{---} \downarrow Q(K_1) \\
 E_1 & \xrightarrow{\quad R \quad} & E_2
 \end{array}$$

Theorem (DF, Marotta, Szabo [4])

Let g_1 be a Riemannian metric on M_1 . If $\mathcal{L}_X g = \mathcal{L}_X F = 0$, $\forall X \in \mathcal{F}_2$, then TFAE

- ① $K_1^\perp \cap \Phi(K_2) \subseteq K_1$.
- ② There exists a unique Riemannian metric g_2 on M_2 such that R is a generalised isometry.

The T-duality relation II

The T-duality relation $R: E \rightarrow E'$ is then the composition of CA relations:

$$\begin{array}{ccc}
 \mathcal{E} & \xrightarrow{\Phi = \varphi_* \circ e^F} & \mathcal{E} \\
 Q(K_2)^T \uparrow \text{dashed} & & \text{dashed} \downarrow Q(K_1) \\
 E_1 & \xrightarrow{\text{dashed } R} & E_2
 \end{array}$$

Theorem (DF, Marotta, Szabo [4])

Let g_1 be a Riemannian metric on M_1 . If $\mathcal{L}_X g = \mathcal{L}_X F = 0$, $\forall X \in \mathcal{F}_2$, then TFAE

- ① $K_1^\perp \cap \Phi(K_2) \subseteq K_1$.
- ② There exists a unique Riemannian metric g_2 on M_2 such that R is a generalised isometry.

Application: T-dualities for Torus bundles

Cavalcanti-Gualtieri [5] formulate the correspondence space picture for T-dual torus bundles over a base B :

Endow M_i with H-flux H_i . Take $\mathcal{M} = M_1 \times_B M_2$ and $q_1: \mathcal{M} \rightarrow M_2$ etc. Then M_1 is T-dual to M_2 if

$$q_2^* H_1 - q_1^* H_2 = dF,$$

with $F: \mathfrak{t}_1 \otimes \mathfrak{t}_2 \rightarrow \mathbb{R}$ invariant and non-degenerate.

The condition $\mathcal{L}_X F = 0$ gives requisite invariance, and item (1) gives non-degeneracy.

The resulting Riemannian metrics g_1, g_2 satisfy the Buscher rules.

Application: T-dualities for Torus bundles

Cavalcanti-Gualtieri [5] formulate the correspondence space picture for T-dual torus bundles over a base B :

Endow M_i with H-flux H_i . Take $\mathcal{M} = M_1 \times_B M_2$ and $q_1: \mathcal{M} \rightarrow M_2$ etc. Then M_1 is T-dual to M_2 if

$$q_2^* H_1 - q_1^* H_2 = dF,$$

with $F: \mathfrak{t}_1 \otimes \mathfrak{t}_2 \rightarrow \mathbb{R}$ invariant and non-degenerate.

The condition $\mathcal{L}_X F = 0$ gives requisite invariance, and item (1) gives non-degeneracy.

The resulting Riemannian metrics g_1, g_2 satisfy the Buscher rules.

Application: T-dualities for Torus bundles

Cavalcanti-Gualtieri [5] formulate the correspondence space picture for T-dual torus bundles over a base B :

Endow M_i with H-flux H_i . Take $\mathcal{M} = M_1 \times_B M_2$ and $q_1: \mathcal{M} \rightarrow M_2$ etc. Then M_1 is T-dual to M_2 if

$$q_2^* H_1 - q_1^* H_2 = dF,$$

with $F: \mathfrak{t}_1 \otimes \mathfrak{t}_2 \rightarrow \mathbb{R}$ invariant and non-degenerate.

The condition $\mathcal{L}_X F = 0$ gives requisite invariance, and item (1) gives non-degeneracy.

The resulting Riemannian metrics g_1, g_2 satisfy the Buscher rules.

Application: Doubled Geometry

Instead of $\mathcal{M}$ being the fibred product space, we can consider it being the simultaneous doubled space of M_1 and M_2 .

From a generalised metric (g_1, b_1) on E_1 , one can form a Riemannian metric G_1 on $\mathcal{M}$.

Let $X_1, \dots, X_N$ be coordinate vector fields on $\mathcal{M}$, and $X_1, \dots, X_n$ generate the foliation $T\mathcal{F}_2$ (with $n = N/2$).

Now pick $k < n$ and a diffeomorphism φ , and define a foliation by $T\mathcal{F}_1$ generated by the following n vector fields:

$$\varphi_*(X_i) \text{ for } i = k + 1, \dots, n, \quad \varphi_*(X_p) \text{ for } p = n + 1, \dots, n + k.$$

Upon forming the T-duality relation $R: E_1 \rightarrow E_2$, one has the following results for the metrics:

$$\varphi^* G_2 = G_1.$$

Application: Doubled Geometry

Instead of $\mathcal{M}$ being the fibred product space, we can consider it being the simultaneous doubled space of M_1 and M_2 .

From a generalised metric (g_1, b_1) on E_1 , one can form a Riemannian metric G_1 on $\mathcal{M}$.

Let $X_1, \dots, X_N$ be coordinate vector fields on $\mathcal{M}$, and $X_1, \dots, X_n$ generate the foliation $\mathcal{TF}_2$ (with $n = N/2$).

Now pick $k < n$ and a diffeomorphism φ , and define a foliation by $\mathcal{TF}_1$ generated by the following n vector fields:

$$\varphi_*(X_i) \text{ for } i = k + 1, \dots, n, \quad \varphi_*(X_p) \text{ for } p = n + 1, \dots, n + k.$$

Upon forming the T-duality relation $R: E_1 \rightarrow E_2$, one has the following results for the metrics:

$$\varphi^* G_2 = G_1.$$

Application: Doubled Geometry

Instead of $\mathcal{M}$ being the fibred product space, we can consider it being the simultaneous doubled space of M_1 and M_2 .

From a generalised metric (g_1, b_1) on E_1 , one can form a Riemannian metric G_1 on $\mathcal{M}$.

Let $X_1, \dots, X_N$ be coordinate vector fields on $\mathcal{M}$, and $X_1, \dots, X_n$ generate the foliation $T\mathcal{F}_2$ (with $n = N/2$).

Now pick $k < n$ and a diffeomorphism φ , and define a foliation by $T\mathcal{F}_1$ generated by the following n vector fields:

$$\varphi_*(X_i) \text{ for } i = k + 1, \dots, n, \quad \varphi_*(X_p) \text{ for } p = n + 1, \dots, n + k.$$

Upon forming the T-duality relation $R: E_1 \rightarrow E_2$, one has the following results for the metrics:

$$\varphi^* G_2 = G_1.$$

Application: Doubled Geometry

Instead of $\mathcal{M}$ being the fibred product space, we can consider it being the simultaneous doubled space of M_1 and M_2 .

From a generalised metric (g_1, b_1) on E_1 , one can form a Riemannian metric G_1 on $\mathcal{M}$.

Let $X_1, \dots, X_N$ be coordinate vector fields on $\mathcal{M}$, and $X_1, \dots, X_n$ generate the foliation $T\mathcal{F}_2$ (with $n = N/2$).

Now pick $k < n$ and a diffeomorphism φ , and define a foliation by $T\mathcal{F}_1$ generated by the following n vector fields:

$$\varphi_*(X_i) \text{ for } i = k + 1, \dots, n, \quad \varphi_*(X_p) \text{ for } p = n + 1, \dots, n + k.$$

Upon forming the T-duality relation $R: E_1 \rightarrow E_2$, one has the following results for the metrics:

$$\varphi^* G_2 = G_1.$$

Application: Doubled Geometry

Instead of $\mathcal{M}$ being the fibred product space, we can consider it being the simultaneous doubled space of M_1 and M_2 .

From a generalised metric (g_1, b_1) on E_1 , one can form a Riemannian metric G_1 on $\mathcal{M}$.

Let $X_1, \dots, X_N$ be coordinate vector fields on $\mathcal{M}$, and $X_1, \dots, X_n$ generate the foliation $T\mathcal{F}_2$ (with $n = N/2$).

Now pick $k < n$ and a diffeomorphism φ , and define a foliation by $T\mathcal{F}_1$ generated by the following n vector fields:

$$\varphi_*(X_i) \text{ for } i = k + 1, \dots, n, \quad \varphi_*(X_p) \text{ for } p = n + 1, \dots, n + k.$$

Upon forming the T-duality relation $R: E_1 \rightarrow E_2$, one has the following results for the metrics:

$$\varphi^* G_2 = G_1.$$

Outlook

- 1 Fit Poisson-Lie T-duality into this framework.
- 2 Add more structure: divergences (or CA connections), allowing to incorporate the dilaton, generalised Ricci tensors and Ricci flow; generalised complex structures, allowing incorporation of branes.

End

Thanks you for listening! Questions?

References: [1] Vysoký <https://arxiv.org/abs/1910.05347>

[2] Bursztyn, Cavalcanti, Gualtieri

<https://arxiv.org/abs/math/0509640>

[3] Zambon <https://arxiv.org/abs/math/0701740>

[4] DF, Marotta, Szabo <https://arxiv.org/abs/2308.15147>

[5] Cavalcanti, Gualtieri <https://arxiv.org/abs/1106.1747>